Rollout recommendation: simplified KYC

Simulated data. Pebble Credit is fictional. Effects were planted in a data generator; this memo demonstrates the analysis method, not a real company's results.

Decision under the pre-registered rule: Iterate

Context

In the 4-week pre-period, the largest funnel drop-off was **KYC start → KYC completed**: 8,763 applicants lost (41.5% of all drop-off; KYC completion rate 49.7% of users who started KYC).

What we tested

A simplified KYC flow vs the current flow, randomized 50/50 at KYC start (8,639 control, 8,740 treatment; sample-ratio check p = 0.44). Analysis is intention-to-treat. The decision rule, metrics and margins were frozen before any data existed.

Results

- **KYC completion rate (primary):** 50.0% → 54.6%, **+4.55 pp** (95% CI [3.06, 6.03] pp; relative lift +9.1%, p = 2.0e-09).
- **Disbursal share of randomized users (secondary):** +2.24 pp (95% CI [0.97, 3.51] pp). The KYC gain survives the credit decision.
- **Early-delinquent loans per 1,000 randomized users (decision guardrail):** 15.51 → 16.93, +1.42 (one-sided 95% bound +4.59 vs margin +2.1): **inconclusive**.
- Diagnostics: early delinquency rate per loan -0.04 pp (bound +1.24 vs +1.5: pass); approval rate +0.64 pp (bound -1.13 vs -3.0: pass).

Why the per-user count rises while the per-loan rate does not: the treatment arm disbursed 251.9 loans per 1,000 randomized users vs 229.5, so it carries more delinquent loans even at an unchanged per-loan rate. Only the per-user count is a randomized comparison.

Decision

- ✓ Primary metric significant (p = 2.0e-09 < 0.05)
- ✓ Primary effect positive (+4.55 pp)
- ✗ Decision guardrail passes non-inferiority (label: inconclusive)

The conversion gain is real, but the data cannot yet rule out an increase in early-delinquent loans beyond the break-even margin. Resolving it at the observed estimate would need about 188,179 randomized users per arm (vs 8,639 now), roughly 86 weeks of KYC traffic. The per-loan early delinquency rate shows no detectable deterioration (-0.04 pp), so the open question is mostly precision: the guardrail's 95% CI spans 7.6 per 1,000, wider than the 2.1 margin. A longer test alone is impractical at current traffic, so the next step is a staged exposure with cohort-level delinquency monitoring while a tighter variant (keeping one check for thin-file applicants) is prepared for a re-test.

Risks

- **Selection effect:** the simpler flow admits marginal applicants with weaker credit profiles.
- **Outcome maturity:** early delinquency covers only the first EMI; later losses are not yet observable.
- **Novelty effect:** a 4-week test may overstate a lift that fades.
- **Simulated data:** results demonstrate the method. The true planted effect (+4.76 pp) lies inside the 95% CI; A/A false-positive rate 4.9%, CI coverage 95.4%.

Rollout plan

- Do not roll out to everyone. Keep a controlled 10% → 50% exposure as an extended test until the decision guardrail resolves.
- Monitor early-delinquent loans per 1,000 KYC entrants and the per-loan rate at 30 / 60 / 90 days after disbursal for each exposure cohort.
- **Rollback triggers:** the early-delinquent loans per 1,000 estimate exceeds the +2.1 margin (guardrail fails) at any monitoring point, or the KYC completion gain's 95% CI falls below zero.

Analysis by Dev Mittal · every figure is computed by the pipeline and read from outputs/.